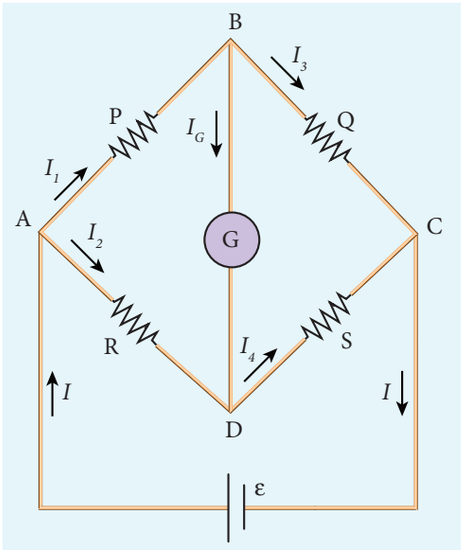


Figure 2.25 Wheatstone's bridge

through the galvanometer is I_G and its resistance is G.

Applying Kirchhoff's current rule to junction B and D respectively.

$$I_1 - I_G - I_3 = 0 \quad (2.45)$$

$$I_2 + I_G - I_4 = 0 \quad (2.46)$$

Applying Kirchhoff's voltage rule to loop $ABDA$,

$$I_1P + I_GG - I_2R = 0 \quad (2.47)$$

Applying Kirchhoff's voltage rule to loop $ABCDA$,

$$I_1P + I_3Q - I_4S - I_2R = 0 \quad (2.48)$$

When the points B and D are at the same potential, the bridge is said to be balanced. As there is no potential difference between B and D, no current flows through galvanometer ($I_G = 0$). Substituting $I_G = 0$ in equation (2.45), (2.46) and (2.47), we get

$$I_1 = I_3 \quad (2.49)$$

$$I_2 = I_4 \quad (2.50)$$

$$I_1P = I_2R \quad (2.51)$$

Using equation (2.51) in equation (2.48)

$$I_3Q = I_4S \quad (2.52)$$

Dividing equation (2.52) by equation (2.51), we get

$$\frac{P}{Q} = \frac{R}{S} \quad (2.53)$$

This is the condition for bridge balance. Only under this condition, galvanometer shows null deflection. Suppose we know the values of two adjacent resistances, the other two resistances can be compared. If three of the resistances are known, the value of unknown resistance (fourth one) can be determined.





A galvanometer is an instrument used for detecting and measuring even very small electric currents. It is extensively useful to compare the potential difference between various parts of the circuit.

EXAMPLE 2.23

In a Wheatstone's bridge P = 100 Ω, Q = 1000 Ω and R = 40 Ω. If the galvanometer shows zero deflection, determine the value of S.

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